Problem Set 0

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Problem 0-1.

(a) $\{6, 12\}$

(b) 7

(c) 3

Problem 0-2.

(a) $E[x] = E[x_1] + E[x_2] + E[x_3]$
$E[x_i] = \frac{1}{2} * 1 + \frac{1}{2} * 0 = \frac{1}{2}$
$E[x] = \frac{3}{2}$

(b) $E[Y] = E[D_1] * E[D_2]$
$E[D_i] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = 3.5$
$E[Y] = 3.5 * 3.5 = 12.25$

(c) $E[X + Y] = E[X] + E[Y] = 1.5 + 12.25 = 13.75$

Problem 0-3.

$A = \frac{600}{6} = 100$
$B = 60 \mod 42 = 18$

(a) True

$$\begin{aligned} 100 &\mod 2 = 0 \\ 18 &\mod 2 = 0 \\ &\implies A \equiv B \pmod 2 \end{aligned}$$

(b) False

$$\begin{aligned} 100 &\mod 3 = 1 \\ 18 &\mod 3 = 0 \\ &\implies A \not\equiv B \pmod 3 \end{aligned}$$

(c) False

$$\begin{aligned} 100 &\mod 4 = 0 \\ 18 &\mod 4 = 2 \\ &\implies A \not\equiv B \pmod 4 \end{aligned}$$

Problem 0-4.
Base case: for $n = 1$,

$$\sum_{i=1}^{1} i^3 = [\frac{1 * (2)}{2}]^2$$
$$1 = 1 \implies true$$

Inductive Step: Suppose the equation is true for $n = k$, prove that it's true for $n = k + 1$

$$\begin{aligned} \sum_{i=1}^{k+1} i^3 &= (k+1)^3 + \sum_{i=1}^{k} i^3 \\ &= (k+1)^3 + \left(\frac{k(k+1)}{2}\right)^2 \\ &= (k+1)^3 + \frac{k^2(k+1)^2}{4} \\ &= \frac{4(k+1)^3 + k^2(k+1)^2}{4} \\ &= \frac{(k+1)^2(k^2+4k+4)}{4} = \left(\frac{(k+1)(k+2)}{2}\right)^2 \square \end{aligned}$$

Problem 0-5.
Base case: $k = 1$. $\implies$ The graph has 1 vertex and $|E| = 0$ edges $\implies$ A graph with one vertex and no edges is trivially acyclic $\implies$ Therefore, the base case holds.

Inductive hypothesis: Assume that every connected graph with k vertices and $k-1$ edges is acyclic.

Inductive step: Consider a connected graph G with $k+1$ vertices and k edges.

Let $\deg(v)$ denote the degree of vertex v. The sum of degrees of all vertices is $2|E| = 2k$.
The average degree is

$$\frac{\sum_{v \in V} \deg(v)}{|V|} = \frac{2k}{k+1} < 2.$$

Since the average degree is less than 2, there exists at least one vertex v with degree 1. Let v be connected to a single vertex u. Remove vertex v and its single edge (v, u). Denote the remaining graph as G_0.
G_0 has k vertices and $k-1$ edges. G_0 is connected because removing a leaf cannot disconnect the graph. By the inductive hypothesis, G_0 is acyclic.
Any cycle in G would have to include v, but $\deg(v) = 1$, so v cannot be part of any cycle.
Hence, G is acyclic □

Problem 0-6.

```
def count_long_subarray(A):
    '''
    Input:  A     | Python Tuple of positive integers
    Output: count | number of longest increasing subarrays of A
    '''
    count = 0
    i = 0
    j = 0
    maxlength = 0
    n = len(A)

    while j < n:
      while j < n - 1 and A[j + 1] > A[j]:
        j += 1

      length = j - i + 1
      if length > maxlength:
        maxlength = length
        count = 1
      elif length == maxlength:
        count += 1

      j += 1
      i = j

    return count
```